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Dale's principle

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In the second half of the nineteenth century, two theories were proposed in relation to the structure and function of the nervous system; the *syncytium theory*, which assumed that neurones were interconnected by protoplasmic bridges and did not act independently, and the *cell theory*, which considered the individual neurones to be independent units. In the first half of this century, the cell theory became more and more widely accepted. This implied that there must be mechanisms for transmission of information from one neurone to another, and, again, two rival theories were proposed, that transmission was electrical or that it was chemical, the latter being first proposed by Elliott in 1904 [7]. In 1921, Otto Loewi [10] was the first to unequivocally demonstrate that chemical transmission occurred when he investigated the mechanism of the action of the vagus nerve on the heart; this was later shown to be due to the release of acetylcholine (ACh) by the nerve terminals. It is now generally accepted that chemical transmission is almost universal in the vertebrate nervous system, but electrical transmission also occurs. The chemical hypothesis of synaptic transmission led to the search for the identification of the transmitter(s) for each neurone.

Highlights of the experiments and ideas that flourished at the beginning of this century are contained in the writings of Henry Dale (see [2]), for several decades a leading figure in British physiology and pharmacology. Dale clarified the action of ACh at synapses in autonomic ganglia and at the neuromuscular junction. He showed that some nerves were cholinergic and others were adrenergic and coined these terms. His findings generated the concept of the chemical identity of each neurone. According to this view, different functional effects on target cells had to be attributed to different chemically identifiable neurones.

In the peripheral nervous system, following a lesion, cholinergic axons can replace each other and adrenergic axons can replace each other, but axons of different chemical types are not interchangeable. Dale's interest in the apparent chemical unity of each neurone was further stimulated in connection with the possibility of chemical transmission in the mammalian CNS. He put forward the hypothesis that where the transmitter at one terminal of an axon can be identified, as he had done for sensory neurones that release a transmitter peripherally to generate antidromic vasodilatation, this transmitter would provide a plausible candidate for the transmitter at the axon's central synapses [1]. This was a stimulus for further investigations.

However, the mere demonstration of different effects being due to chemically different neurones was not a proof that each neurone was secreting only one transmitter or that each neurone was secreting the same transmitter(s) at all its terminals. Indeed, Dale was well aware of the possibility that more than one transmitter might be released by a neurone, since he had observed an atropine-resistant effect of the vagus nerve on the gut and an atropine-resistant parasympathetic vasodilatation; he suggested that

"We may suppose that vagus effects not paralysed by atropine are not humorally transmitted at all, or that the transmitter is not a choline ester, but in the latter case, we shall have to postulate not one, but several other transmitters with different degrees of liability to the antagonism of atropine." [3]

Subsequently, Dale's hypothesis was promoted by Eccles, who referred to it as "Dale's Principle" and Eccles stated that:

"the same chemical transmitter is released from all the synaptic terminals of a neurone," [6]

and

"it will readily be appreciated that the same mechanical manufacturing goes on throughout the whole extent of a cell, and that a cell cannot make one kind of transmitter substance for some of its terminals and another kind for others of its terminals." [4]

While the term "transmitter" is used in the singular, the statements were intended to imply the chemical unity of the neurone rather than to imply that only one transmitter could be produced. However, these remarks were widely interpreted to mean that, according to Dale's Principle, each type of neurone only released a single transmitter. As Eccles later said:

"I proposed that Dale's Principle be defined as stating that at all the axonal branches of a neurone, there was liberation of the same transmitter substance or substances." [5]

As Dale himself said in 1954 in a letter to Eccles (see [8], p. 53)

"One must suppose, I think, that there is a constant drift of transmitter, with its associated enzymes, from cell body down the axon, so that there is an accumulation at the ending. . . . "

It seems clear that Dale himself did not elevate his proposal to the level of a principle, especially in relation to a neurone secreting only a single transmitter, but retained it as a hypothesis that, once the nature of the transmitter(s) at one terminal of a neurone had been identified, gave a hint as to the likely nature of the transmitter(s) at other terminals. In view of the recent findings by Jonas et al. [9], we now know that two classical inhibitory transmitters,

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GABA and glycine, can coexist in the same synaptic vesicle, a fact that is not in conflict with Dale's ideas.

REFERENCES

1. Dale, H. H. Pharmacology and nerve-endings. Proc. R. Soc. Med. 28:319–332; 1935.
2. Dale, H. H. Adventures in physiology. London: Pergamon Press; 1953.
3. Dale, H. H.; Gaddum, J. H. Reactions to denervated voluntary muscle and their bearing on the mode of action of parasympathetic and related nerves. J. Physiol. (Lond.) 70:109–144; 1930.
4. Eccles, J. C. (Guest Editor). The clinical significance of research work on the chemical transmitter substances of the nervous system. Med. J. Aust. June 1:747–753; 1957.
5. Eccles, J. C. From electrical to chemical transmission in the central nervous system. R. Soc. London Notes and Records 30:219–230; 1976.
6. Eccles, J. C.; Fatt, P.; Koketsu, K. Cholinergic and inhibitory synapses in a pathway from motor-axon collaterals to motoneurones. J. Physiol. (Lond.) 126:524–562; 1954.
7. Elliot, T. R. On the action of adrenalin. J. Physiol. (Lond.) 31:xx–xxi; 1904.
8. Girolami, P.; Táboríková, H.; Nisticò, G. In memory of Sir Henry Dale. Rome: Accademia Romana di Scienze Biologiche e Mediche; 1994.
9. Jonas, P.; Bischofberger, J.; Sandkuhler, J. Corelease of two fast neurotransmitters at a central synapse. Science 281:419–424; 1998.
10. Loewi, O. Über humorale Übertragbarkeit des Herznervenwirkung. Pflügers Arch. 189:239–242; 1921.